

Corporate Learning Management System (LMS) SaaS

Functional Workflow & System Architecture

1. Overall SaaS Structure

The proposed solution will be a **multi-tenant Corporate Learning Management System (LMS) delivered as a SaaS platform**.

Organizations will subscribe to the platform and receive their own secure workspace where their administrators, HR teams, managers, trainers, and employees can manage and participate in corporate learning activities.

Each organization will operate independently within the platform.

System Structure

Your SaaS Platform

→ Company A

→ Administrator / HR

→ Managers

→ Employees

→ Company B

→ Administrator / HR

→ Managers

→ Employees

→ Company C

→ Administrator / HR

→ Managers

→ Employees

Each company's information will remain securely isolated. Employees, courses, reports, documents, assessments, and other organizational information belonging to one company will not be accessible by another company.

2. Platform Super Administrator

The platform owner will have access to a centralized **Super Administration Panel** from which the entire SaaS ecosystem can be managed.

The Super Administrator will be responsible for managing:

Module	Purpose
Companies	Manage registered organizations
Subscriptions	Manage monthly and annual subscriptions
Billing	Manage payments, invoices, and renewals
Users	Manage company administrators
Usage	Monitor employees, courses, and storage utilization
Subscription Plans	Configure Starter, Business, Professional, and Enterprise plans
Global Courses	Maintain an optional centralized training library
Support	Manage customer support requests
Analytics	Monitor active companies, users, subscriptions, and revenue
Platform Settings	Manage email, integrations, storage, and system configuration

Super Administrator Workflow

Super Administrator

- Company Registration / Approval
- Subscription Activation
- Company Workspace Creation
- Company Administrator Invitation
- Organization Begins Using the Platform

3. Company Registration & Subscription

Organizations will be able to register directly through the SaaS platform.

Registration Workflow

Get Started

- Company Registration
- Enter Company Information
- Select Subscription Plan
- Complete Payment
- Company Workspace Created
- Administrator Account Created

Company Information

The onboarding process may capture:

- Company Name
- Industry
- Country

- Number of Employees
- Administrator Name
- Administrator Email
- Phone Number
- Selected Subscription Plan
- Billing Information

Each organization can be provided with a dedicated workspace such as:

`acme.yourlms.com`

or

`yourlms.com/company/acme`

Enterprise customers may also be provided with a custom domain such as:

`training.acme.com`

Custom domain functionality may be included as an Enterprise feature.

4. Company Administrator / HR Dashboard

The **Company Administrator / HR Dashboard** will serve as the primary management interface for each organization.

HR and authorized administrators will be able to control employee training activities from this dashboard.

The dashboard may display:

- Total Employees
- Active Employees
- Total Courses
- Assigned Courses
- Completed Courses
- Pending Training
- Overdue Training
- Certificates Issued
- Upcoming Deadlines
- Failed Assessments
- Department Performance
- Overall Completion Rate
- Compliance Percentage

Example Dashboard

Employees: 840

Courses: 47

Course Assignments: 3,650
Completed: 2,920
Pending: 610
Overdue: 120
Completion Rate: 80%

This provides HR management with an immediate overview of the organization's training and compliance status.

5. Organizational Structure

The LMS should support the internal organizational hierarchy of each company rather than maintaining only a basic employee list.

A typical structure may include:

Company
→ Country / Location
→ Department
→ Team
→ Designation
→ Employee

For example:

Organization
→ India
→ Marketing
→ Design
→ Production
→ Operations

→ Germany
→ Sales
→ Operations

→ United States
→ Sales
→ Production

This structure will allow administrators to assign training at different organizational levels.

Courses may be assigned to:

- All Employees
- Individual Employees
- Specific Departments

- Managers
 - Locations
 - Designations
 - Teams
 - Employee Groups
-

6. Employee Management

The Employee Management module will enable Company Administrators and HR personnel to centrally manage the organization's workforce.

Employees may be added manually or imported in bulk through CSV or Excel files.

Employee Information

The system may store:

- Employee ID
- Full Name
- Email Address
- Phone Number
- Department
- Designation
- Reporting Manager
- Location
- Joining Date
- Employment Type
- Employment Status

Administrative Actions

Authorized administrators may:

- Add Employees
- Edit Employee Information
- Bulk Import Employees
- Deactivate Employees
- Transfer Employees Between Departments
- Assign Managers
- Reset Passwords
- Assign Courses
- Review Training History
- Review Certificates
- Monitor Compliance Status

Future integrations may include Microsoft Entra ID, Google Workspace, HRMS platforms, or other identity management systems.

7. Course Creation Workflow

The Course Management module will allow authorized administrators and training managers to create structured corporate learning programs.

Course Creation Process

Create Course

- Enter Basic Information
- Create Course Sections
- Add Lessons
- Upload Learning Content
- Configure Assessment
- Define Completion Rules
- Configure Certificate
- Publish Course

Example Course

Workplace Safety Training

Module 1 - Introduction

- Video

Module 2 - Safety Procedures

- PDF
- Video
- Images

Module 3 - Emergency Procedures

- Video

Module 4 - Assessment

- 20 Questions

Final Stage

- Certificate of Completion
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8. Supported Content Types

The course management system should support multiple forms of learning content.

Supported formats may include:

- Video
- PDF
- PowerPoint Presentation
- Images
- Text-Based Lessons
- Audio
- External Video Links
- Web Links
- Quizzes
- Assignments
- Surveys
- SCORM Packages
- xAPI Content

SCORM and xAPI support can be particularly valuable for enterprise customers using externally developed corporate training packages.

9. Course Builder

The LMS should provide an intuitive course builder that enables administrators to organize lessons into sections and modules.

Example

Course: Fire Safety

Section 1 - Introduction

- Lesson 1: Video
- Lesson 2: PDF

Section 2 - Fire Prevention

- Lesson 3: Video
- Lesson 4: Article

Section 3 - Assessment

- Quiz: 15 Questions

A drag-and-drop course builder may be implemented to make course administration easier and more efficient.

10. Course Configuration & Completion Rules

Administrators should be able to configure the learning and completion requirements for every course.

Configuration options may include:

- Mandatory or Optional Course
- Passing Percentage
- Maximum Assessment Attempts
- Certificate Availability
- Certificate Validity
- Course Completion Deadline
- Sequential Lesson Progression
- Lesson Skipping Permission
- Minimum Video Completion Percentage
- Assessment Requirement

Example

Course Type: Mandatory
Passing Score: 80%
Maximum Attempts: 3
Certificate: Enabled
Certificate Validity: 12 Months
Course Deadline: 30 Days
Lesson Sequence: Sequential
Allow Lesson Skipping: No
Video Completion Requirement: 90%
Assessment Required: Yes

These settings are particularly important for compliance-based corporate training.

11. Course Assignment

Once a course has been published, administrators can assign it to selected employees or organizational groups.

Example

Course: Cybersecurity Awareness

Assigned To:

- IT Department
- Finance Department

Deadline: 30 September
Mandatory: Yes
Reminder Frequency: Every 3 Days
Escalation: Notify Manager After Deadline

The platform will automatically enroll all eligible employees based on the selected assignment criteria.

12. Learning Paths

The system should support **Learning Paths**, allowing multiple courses to be grouped into structured training journeys.

Example: New Employee Onboarding

1. Welcome to the Organization
2. Company Policies
3. Workplace Safety
4. Cybersecurity Awareness
5. Role-Specific Training
6. Final Assessment

Administrators may configure learning paths so employees must complete courses in a defined sequence.

Learning paths may be used for:

- Employee Onboarding
 - Leadership Development
 - Compliance Programs
 - Role-Based Training
 - Departmental Training
 - Management Development
 - Technical Certification Programs
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13. Automatic Course Assignment

The platform should support rule-based course assignment to minimize repetitive administrative work.

Example Rule 1

If:

Department = Production
AND
Employee = Newly Joined

Then automatically assign:

- Workplace Safety
- Machine Safety
- Emergency Response Training

Example Rule 2

If:

Designation = Manager

Then automatically assign:

- Leadership Training
- Workplace Compliance Training
- Performance Management Training

This automation significantly reduces the administrative workload of HR teams.

14. Employee Learning Portal

Employees will have access to a simple, personalized learning dashboard.

The employee portal may include:

- My Courses
- Mandatory Training
- Continue Learning
- Upcoming Deadlines
- Completed Courses
- Certificates
- Learning History
- Assessment Results
- Training Calendar

Example

Workplace Safety

Progress: 80%

Action: Continue Learning

Cybersecurity Awareness

Progress: 50%

Action: Continue Learning

Company Policies

Progress: 100%

Action: View Certificate

Employees will only have access to learning-related features and will not have access to administrative functionality.

15. Learning Experience & Progress Tracking

When an employee opens a course, the system will guide them through the defined learning journey.

Course

- Lesson 1
- Video
- Lesson Completion
- Lesson 2
- PDF
- Quiz
- Pass Assessment
- Course Completion
- Certificate

The platform should track:

- Course Start Date
- Time Spent
- Video Completion Percentage
- Lessons Completed
- Quiz Score
- Number of Attempts
- Completion Date
- Certificate Status

This data will be available to administrators for reporting and compliance monitoring.

16. Quiz & Assessment Management

The LMS should include a complete assessment management system.

Supported question types may include:

- Multiple Choice
- Multiple Select
- True / False
- Fill in the Blank
- Short Answer
- Long Answer
- Image-Based Questions

A centralized Question Bank should also be available.

Example

Cybersecurity Question Bank: 100 Questions

Assessment Configuration:

Random Questions Per Attempt: 20

Passing Percentage: 80%

Randomized assessments allow different employees to receive different combinations of questions while maintaining the same assessment standard.

17. Certificate Management

Upon successful completion of eligible courses, employees may automatically receive a digital certificate.

A certificate may contain:

- Employee Name
- Course Name
- Assessment Score
- Completion Date
- Certificate Number
- Certificate Validity
- Organization Name

Example

Certificate of Completion

Employee: Rahul Sharma

Course: Workplace Safety Training

Score: 92%

Completion Date: 8 September 2026

Certificate ID: TRN-2026-89272

Certificate validity may be configured as:

- 6 Months
- 1 Year
- 2 Years
- Lifetime

QR-based certificate verification may also be implemented as an advanced feature.

18. Compliance Training Management

The LMS can be positioned not only as a learning platform but also as a comprehensive **Employee Training & Compliance Management System**.

Organizations may use the platform to manage training such as:

- Workplace Safety
- Prevention of Sexual Harassment (POSH)
- Cybersecurity

- GDPR
- Fire Safety
- First Aid
- Code of Conduct
- Data Protection
- Anti-Bribery
- Quality Standards
- Machine Safety
- Internal Policies

Compliance Dashboard

Compliant: 82%

Expiring: 11%

Expired: 7%

This enables management to identify compliance gaps immediately.

19. Course Expiry & Recertification

Certain corporate courses and certifications require periodic renewal.

The LMS should support configurable certificate validity and automatic recertification workflows.

Example

Course: Fire Safety Training

Certificate Validity: 12 Months

Employee Completion Date: 10 September 2026

Certificate Expiry Date: 10 September 2027

Automated Reminder Workflow

30 Days Before Expiry

→ Reminder Notification

15 Days Before Expiry

→ Reminder Notification

7 Days Before Expiry

→ Final Reminder

Certificate Expired

→ Training Automatically Reassigned

This functionality is especially valuable for industrial, manufacturing, healthcare, and compliance-driven organizations.

20. Notification Management

The platform should include a centralized notification engine.

Notifications may be delivered through:

- Email
- In-App Notifications
- Microsoft Teams
- Slack
- WhatsApp, where integrated

Notifications may be triggered by:

- Course Assignment
 - Upcoming Deadline
 - Course Overdue
 - Assessment Failure
 - Certificate Issued
 - Certificate Expiring
 - New Course Published
 - Learning Path Assigned
 - Training Session Scheduled
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21. Manager Portal

Managers should have access to a dedicated dashboard showing only employees within their reporting structure.

Example

Team Size: 20 Employees

Training Completed: 17

Pending: 2

Overdue: 1

Managers may be allowed to:

- View Team Members
- Review Training Progress
- View Pending Courses
- View Overdue Training
- Review Employee Training History
- View Team Reports
- Monitor Compliance Status

Managers will not have access to organization-wide administrative controls unless additional permissions are explicitly assigned.

22. Reporting & Analytics

Comprehensive reporting is essential for corporate LMS customers.

The platform should provide reports such as:

- Course Completion Report
- Employee Training Report
- Department Performance Report
- Assessment Report
- Compliance Report
- Overdue Training Report
- Certificate Expiry Report
- Learning Hours Report
- Manager-Wise Report
- Location-Wise Report
- Course Performance Report
- Employee Progress Report

Reports may be exported to Excel and PDF.

Example

Employee	Department	Course	Score	Status
Amit	Production	Fire Safety	92%	Completed
Rahul	Production	Fire Safety	76%	Failed
Priya	HR	POSH	95%	Completed
Neha	Sales	Cybersecurity	-	Pending

23. HR Dashboard & Training Analytics

HR teams should receive a high-level view of organizational learning and compliance.

Example Metrics

Training Compliance: 87%

Completed Training: 1,420

Pending Training: 156

Overdue Training: 43

Certificates Expiring Soon: 28

Department Performance

Production: 91%
HR: 96%
Sales: 78%
Marketing: 86%
Operations: 82%

This information will allow HR teams to identify departments requiring additional attention.

24. SaaS Subscription Management

Since the LMS will operate as a SaaS platform, organizations can be billed according to their subscription plan and employee capacity.

Example Plans

Plan	Employee Capacity	Core Features
Starter	Up to 50	Basic LMS
Business	Up to 250	LMS + Reporting
Professional	Up to 1,000	Automation + Advanced Features
Enterprise	Custom / Unlimited	SSO + Custom Domain + Enterprise Features

Subscription pricing may also be calculated on a per-user basis.

For example:

Price Per Employee: INR 60 / Month
Employees: 500
Monthly Subscription: INR 30,000

Annual billing plans and volume-based discounts may also be configured.

25. SaaS Billing Workflow

The subscription lifecycle may operate as follows:

Company Registration
→ Select Subscription Plan
→ Complete Payment
→ Subscription Activated
→ Employee Limit Allocated

- Usage Monitoring
- Renewal

If a subscription expires:

- Subscription Expired
 - Grace Period
 - Payment Reminder
 - Account Changed to Read-Only Mode
 - Account Suspension if Payment Remains Outstanding

Organizational data should not be deleted immediately after subscription expiration. Appropriate data-retention policies should be implemented.

26. Multi-Tenant Architecture

The SaaS platform will follow a multi-tenant architecture.

Every company will operate as an independent tenant within the system.

Core business records will maintain a tenant or company relationship through fields such as:

`tenant_id`

or

`company_id`

This relationship will be applied to modules such as:

- Users
- Employees
- Courses
- Assignments
- Assessments
- Certificates
- Reports
- Departments
- Locations
- Learning Paths
- Notifications

Example

Courses Table:

- id
- company_id

- title
- status
- created_by

The system must ensure that:

Company A can access only Company A's data.

Company B can access only Company B's data.

Tenant-level authorization and database filtering will enforce secure data isolation.

27. Roles & Permissions

The platform should use a Role-Based Access Control (RBAC) model.

Recommended roles include:

- Super Administrator
- Company Administrator / HR
- Training Manager
- Department Manager
- Instructor
- Employee

Example Permissions

Company Administrator / HR

- Create Courses
- Manage Employees
- Assign Courses
- Access Reports
- Manage Departments
- Manage Certificates
- Configure Training Rules

Manager

- View Team Members
- Monitor Team Training
- Access Team Reports
- View Compliance Status

Employee

- Access Assigned Courses
- Complete Lessons
- Attempt Assessments
- Download Certificates
- View Learning History

Each organization may configure permissions according to its internal policies.

28. Employee Onboarding Automation

Employee onboarding can be automated through role, department, location, or designation-based rules.

Example Employee

Employee: Amit Sharma
Department: Production
Location: Delhi
Designation: Engineer

The system identifies:

New Employee

- Production Department
- Delhi Location
- Engineer Designation

The LMS may automatically assign:

- Company Orientation
- Code of Conduct
- Workplace Compliance Training
- Fire Safety
- Machine Safety
- Data Security

This removes the requirement for HR to manually assign every onboarding course.

29. Training Calendar

The LMS should provide an organizational training calendar containing both online and offline learning activities.

Example

September 2026

08 September - Fire Safety
10 September - Cybersecurity
12 September - Workplace Compliance
15 September - Machine Training
20 September - Leadership Training

The calendar may support:

- Online Training
 - Classroom Training
 - Webinars
 - Workshops
 - Instructor-Led Training
 - Assessment Dates
 - Certification Events
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30. Classroom & Offline Training

Corporate and industrial organizations frequently conduct physical classroom training in addition to online courses.

The platform should therefore support both:

Online Learning

and

Offline / Classroom Training

Example

Course: Forklift Safety Training
Training Type: Classroom
Trainer: Safety Manager
Location: Plant 2
Date: 15 September
Attendance: 43 Employees

The trainer or administrator may record attendance directly within the LMS.

Employees who successfully complete offline training may also receive certificates through the same system.

31. Employee Training Matrix

The Employee Training Matrix will provide HR and compliance teams with a consolidated view of employee qualifications and training status.

Example

Employee	Fire Safety	Machine Safety	First Aid	Cybersecurity
Amit	Valid	Valid	Expiring	Valid
Rahul	Valid	Missing	Valid	Valid
Priya	Expiring	Valid	Valid	Valid

Training status may be represented as:

- Valid
- Expiring Soon
- Missing
- Expired
- Pending

This feature is particularly valuable for manufacturing, industrial, construction, healthcare, logistics, engineering, and other compliance-intensive organizations.

32. Complete System Workflow

The overall platform workflow will be:

Super Administrator
→ Company Registration
→ Subscription Activation
→ Company Workspace Creation
→ Company Administrator Setup
→ Organization Structure Configuration
→ Employee Import / Creation
→ Course Creation
→ Content Upload
→ Assessment Configuration
→ Certificate Configuration
→ Course Publishing
→ Course Assignment
→ Employee Notification
→ Employee Login
→ Learning & Course Completion
→ Assessment
→ Successful Completion
→ Certificate Generation
→ HR Reporting
→ Compliance Monitoring
→ Certificate Expiry Monitoring
→ Automatic Recertification / Retraining

33. Core Platform Modules

The complete SaaS platform should include the following modules:

1. Dashboard
2. Companies
3. Subscription & Billing
4. Users
5. Employees
6. Departments
7. Locations
8. Designations
9. Courses
10. Course Builder
11. Learning Paths
12. Content Library
13. Quiz & Assessment
14. Question Bank
15. Course Assignment
16. Training Calendar
17. Classroom Training
18. Certificates
19. Employee Training Matrix
20. Reports
21. Notifications
22. Automation Rules
23. Roles & Permissions
24. Integrations
25. Platform Settings

Additional supporting functionality will include:

- Company Registration
 - Tenant Management
 - HR Dashboard
 - Manager Dashboard
 - Employee Portal
 - Compliance Management
 - Course Expiry
 - Recertification
 - Employee Onboarding Automation
 - Progress Tracking
 - Subscription Management
 - Usage Monitoring
 - Data Isolation
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34. Recommended Technology Stack

The application will be developed using a modern, scalable technology architecture.

Backend

Laravel

Laravel will be used for:

- Application Logic
- Authentication
- APIs
- Role & Permission Management
- Multi-Tenant Architecture
- Course Management
- Employee Management
- Assessment Engine
- Notifications
- Reporting
- Subscription Management
- Background Jobs

Database

MySQL

MySQL will be used for securely managing:

- Companies
- Users
- Employees
- Departments
- Courses
- Lessons
- Assessments
- Questions
- Course Assignments
- Training Progress
- Certificates
- Subscriptions
- Notifications
- Reports

Supporting Services

Depending on project requirements, the application may integrate with:

- Cloud File Storage
- Email Services

- WhatsApp API
- Payment Gateways
- Video Hosting Platforms
- Microsoft Teams
- Slack
- HRMS Systems
- Single Sign-On Providers
- Custom Domains

Third-party integrations will be implemented according to the selected service providers and available APIs.

35. Recommended Product Positioning

Instead of positioning the platform only as a traditional Learning Management System, the product can be positioned as a:

Corporate Employee Training & Compliance Management Platform

This positioning communicates a stronger business value because corporate organizations require more than online course delivery.

The platform will help organizations answer critical operational questions such as:

- Which employees have completed mandatory training?
- Which employees have overdue training?
- Which certifications are approaching expiry?
- Which employees require retraining?
- Which departments have compliance gaps?
- Can the organization provide training evidence during an audit?
- Are newly joined employees completing mandatory onboarding?
- Which managers have employees with overdue training?

The system therefore becomes a centralized platform for managing the complete employee learning, certification, compliance, onboarding, and training lifecycle.

36. Business Value

The proposed platform will help organizations:

- Centralize employee learning and training
- Reduce manual HR administration
- Automate mandatory course assignments
- Monitor employee compliance
- Maintain employee training records

- Generate certificates automatically
- Track certification expiry
- Automate retraining
- Support online and classroom training
- Improve audit readiness
- Provide management-level training analytics
- Standardize onboarding processes
- Reduce compliance risks
- Maintain a complete digital training history for every employee

The result will be a scalable SaaS solution that organizations can use as their central system for **Corporate Learning, Employee Training, Certification, and Compliance Management.**